IMEN574 / IDSC574 · PROGRAMMING FOR AX

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Dimensionality Reduction

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Where We Are

01 Programming with AI

Lectures 2–4

02 **Python and Classical Machine Learning**

Lectures 5–13

03 Deep Learning and Foundation Models

Lectures 14–22

Outline: Dimensionality Reduction

01 **Linear Compression**

Preserve reconstruction variance

02 **Learned Representations**

Change the objective

03 **Neighborhood Displays**

Interpret local geometry

01 · Linear Compression

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Representation Objectives

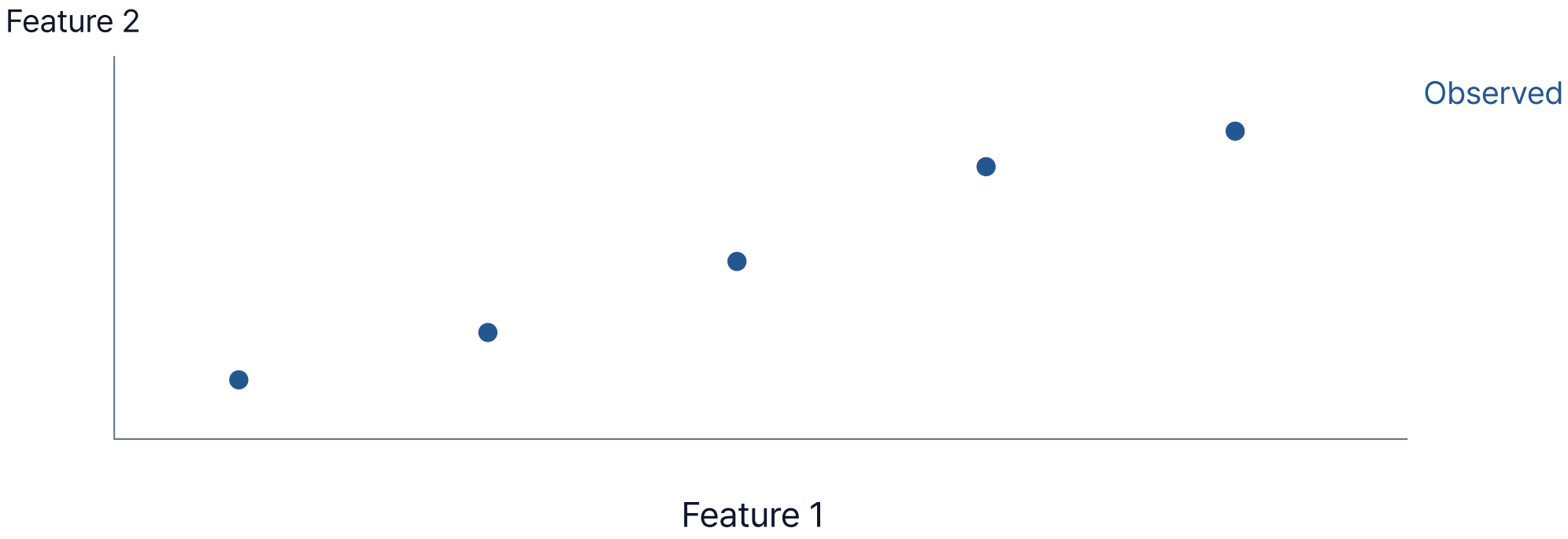
Compression, separation, and visualization ask different questions

Goal	Uses Labels?	Representative Method
Reconstruct input	No	PCA, autoencoder
Separate known classes	Yes	LDA
Show local neighborhoods	No	t-SNE, UMAP

A two-dimensional plot is only one representation objective

Why Fewer Coordinates

Redundant measurements can describe one direction



The five points mostly vary along one direction

Centering the Data

Subtract the training mean before estimating directions

Centered x_1	Centered x_2
-2.0	-1.06
-1.0	-0.66
0.0	-0.06
1.0	0.74
2.0	1.04

Each centered feature has mean zero

Covariance Structure

Off-diagonal terms record joint variation

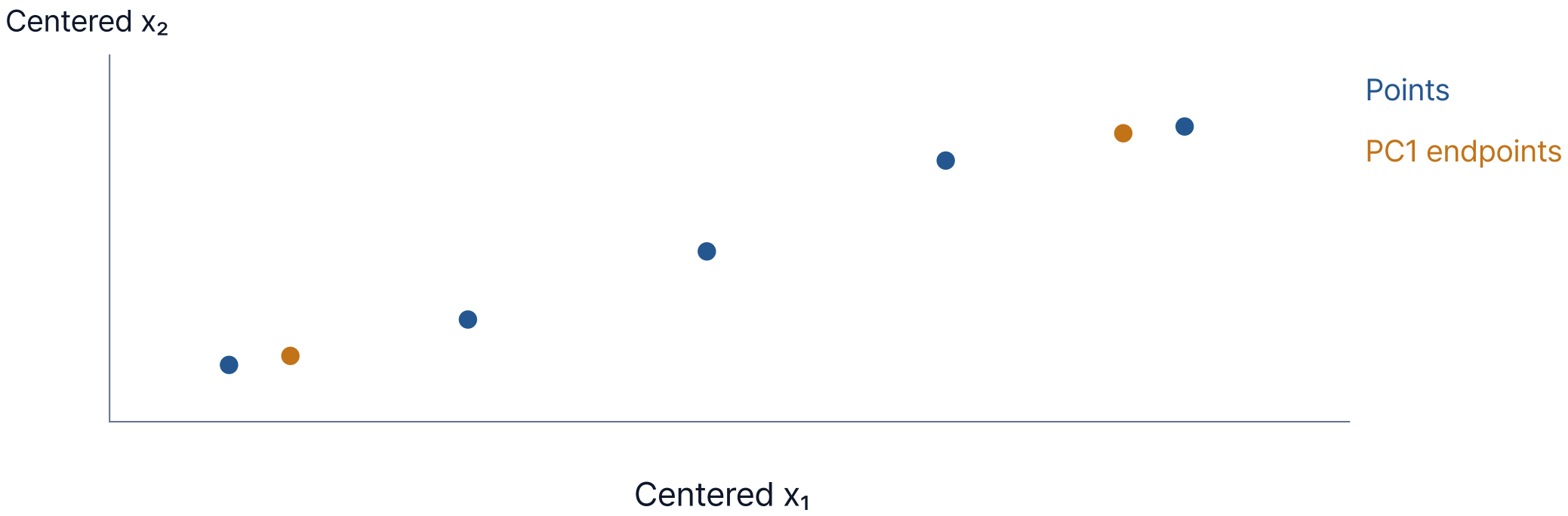
	x_1	x_2
x_1	2.5	1.4
x_2	1.4	0.8

$$C = X^T X / (n - 1)$$

The positive off-diagonal terms indicate shared movement

Principal Component Direction

Maximize projected variance with a unit direction



$$\max \|Xv\|^2 \text{ subject to } \|v\| = 1$$

The first direction follows the largest spread

Projection Scores

Each observation becomes a coordinate along the direction

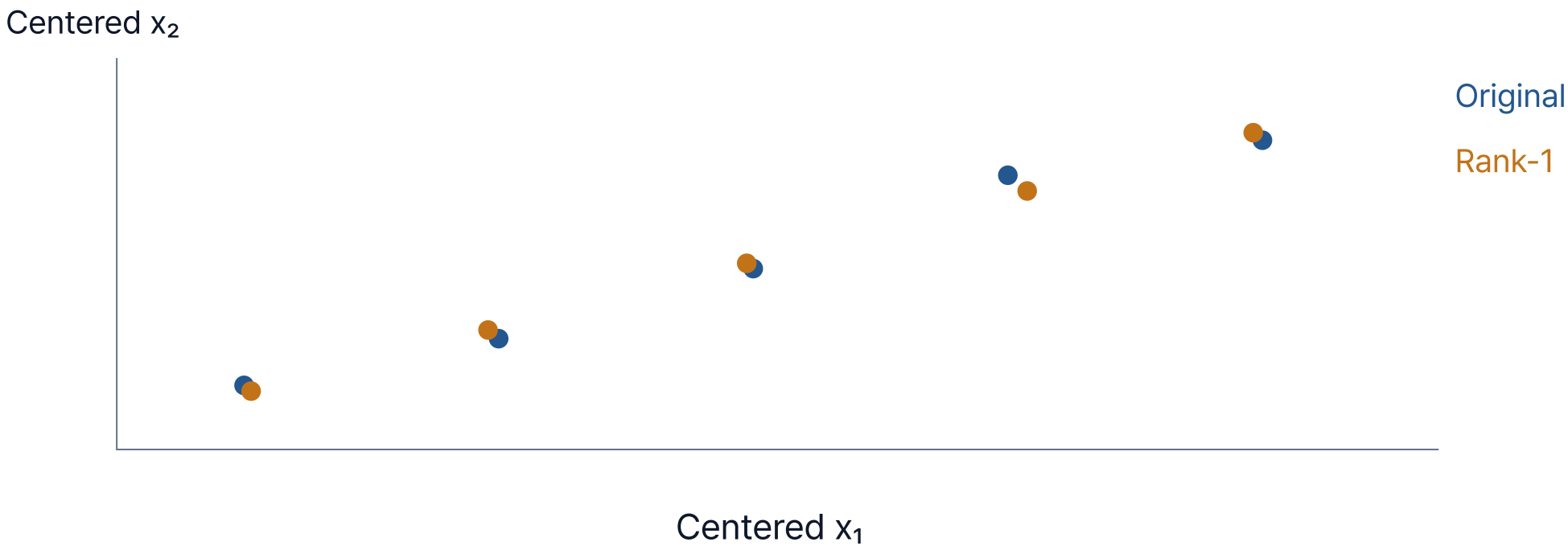
PC1 score
-2.263
-1.195
-0.029
1.234
2.253

$$Z = XV_k$$

One coordinate replaces each two-feature row

Low-Rank Reconstruction

Map reduced coordinates back into the feature space

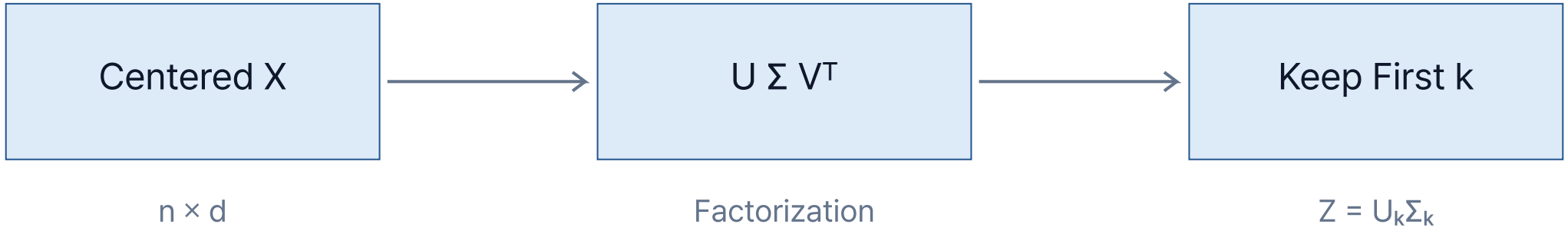


$$\hat{X} = ZV_k^T$$

Reconstruction removes variation orthogonal to PC1

Singular Value Decomposition

SVD computes orthogonal directions without forming covariance



The right singular vectors define the feature directions

PCA With NumPy

Use SVD and verify reconstruction directly

```
X=np.array([[-2,-1],[-1,-.6],[0,0],[1,.8],
[2,1.1]])
X=X-X.mean(0)
U,s,Vt=np.linalg.svd(X,full_matrices=False)
Z=X@Vt[:1].T
print("score shape", Z.shape)
print("MSE", round(np.mean((X-
Z@Vt[:1])**2),5))
```

```
score shape (5, 1)
MSE 0.00426
```

Explained Variance

Choose retained variance rather than only a convenient dimension



Fraction of total variance

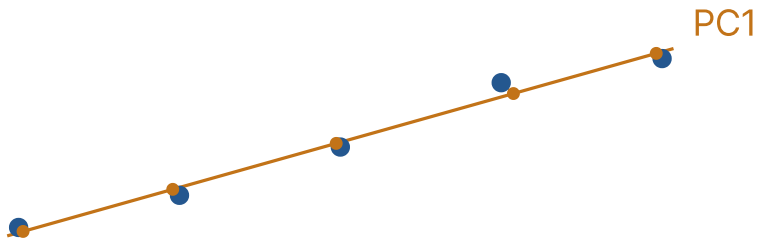
The first component explains most variance in this input

Demo: PCA Projection and Reconstruction

Compute principal axes, scores and reconstruction from the same data

Change retained components from 1 to 2 Scale feature 2 by 100; inspect the covariance and first axis

Feature 2 scale 1 Components 1



Sample covariance (n−1)

2.5 1.4
1.4 0.798

First axis 0.8716, 0.4902

MSE averages all 10 coordinates

Eigenvalues [3.287353, 0.010647]; PC1 variance ratio 0.996772
Scores [-2.263, -1.195, -0.029, 1.234, 2.253]; reconstruction MSE 0.00425875

Answer: PCA Projection and Reconstruction

The same input produces these calculated values

Input / quantity	Result
X=[[-2,-1],[-1,-.6],[0,0],[1,.8],[2,1.1]]	mean=[0,.06]
sample covariance	[[2.5,1.4],[1.4,.798]]
k1 / k2 reconstruction	MSE 0.004258745 / ≈ 0

Caveat: PCA Projection and Reconstruction

PCA preserves variance in the chosen measurement units

Condition	Consequence
Multiply one feature by100	The learned axis changes
Fit a new basis on held-out rows	The evaluation transformation is no longer frozen

Caveat: Feature Scaling

Units change which direction has the most variance

	Feature A scale	Feature B scale
Raw Units	1	100
Standardized	1	1

A hundredfold scale difference can dominate variance

02 · Learned Representations

01 **Linear Compression**

Preserve reconstruction variance

02 **Learned Representations**

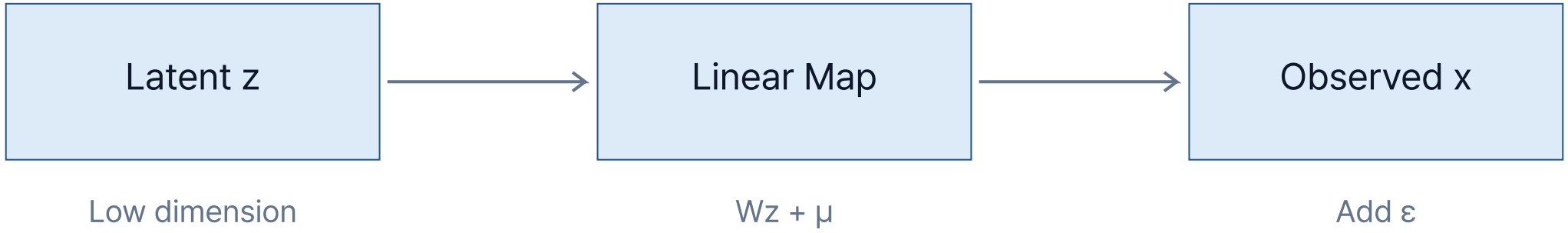
Change the objective

03 **Neighborhood Displays**

Interpret local geometry

Probabilistic PCA

A latent linear model explains covariance plus isotropic noise



The model assigns a density rather than only a projection

Factor Analysis

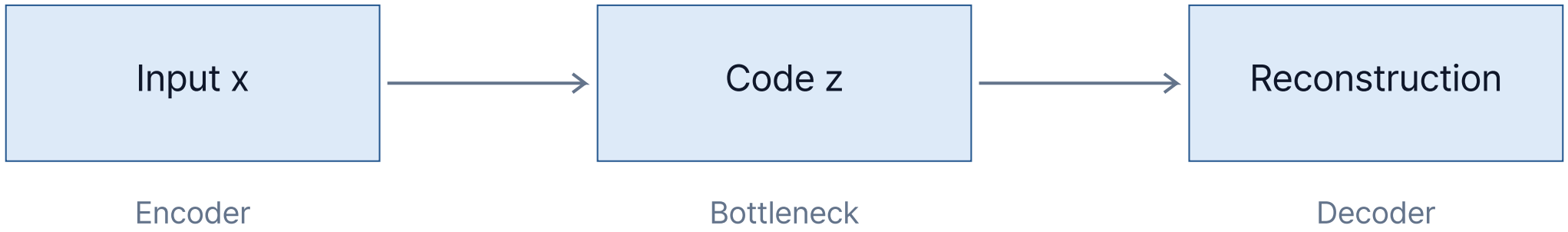
Allow each observed feature its own noise level

	ε_1	ε_2	ε_3
Feature 1	0.2	0	0
Feature 2	0	1.5	0
Feature 3	0	0	0.4

Diagonal noise differs across the measured features

Autoencoder Objective

Learn an encoder and decoder by reconstruction



Nonlinearity expands the possible reconstruction geometry

Linear Autoencoders

A linear bottleneck shares a PCA reconstruction subspace

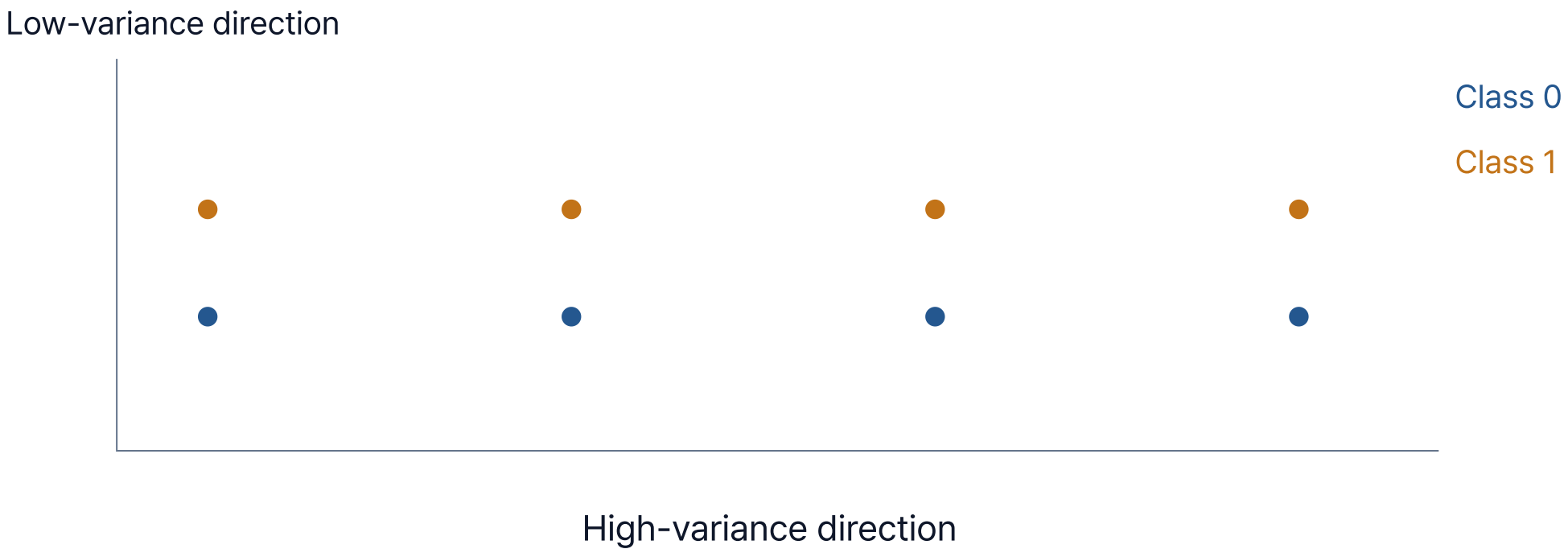
Shape	Operation
$2 \to 1$	Encoder
$1 \to 2$	Decoder

$$\min \|X - XW_eW^d\|^2$$

The reconstruction target connects the two linear models

Supervised Projection

Labels can identify a low-variance separating direction



The small vertical difference separates the classes

Linear Discriminant Analysis

Seek large class separation relative to within-class spread

	Between-Class	Within-Class
Direction A	8	1
Direction B	1	8

$$\text{Maximize between-class / within-class variation}$$

The direction with greater between-class separation is preferred

PCA and LDA Axes

Unsupervised variance and label separation can disagree

Property	PCA	LDA Projection
Labels	Unused	Required
Objective	Total variance	Class separation
Max components	min(n,d)	At most C−1

The label information changes the objective and dimensional limit

Feature Selection

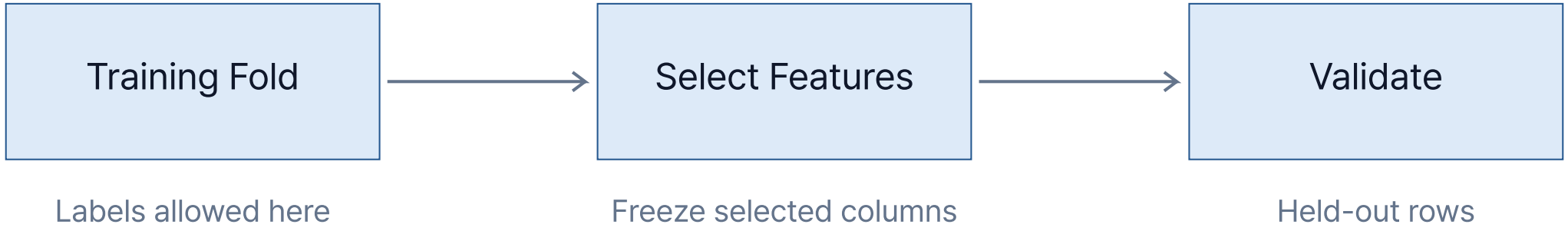
Keep original columns instead of mixing them

	x_1	x_2	x_3	x_4
Columns	Keep	Drop	Keep	Drop

Selected columns remain in their original coordinates

Selection Inside Validation

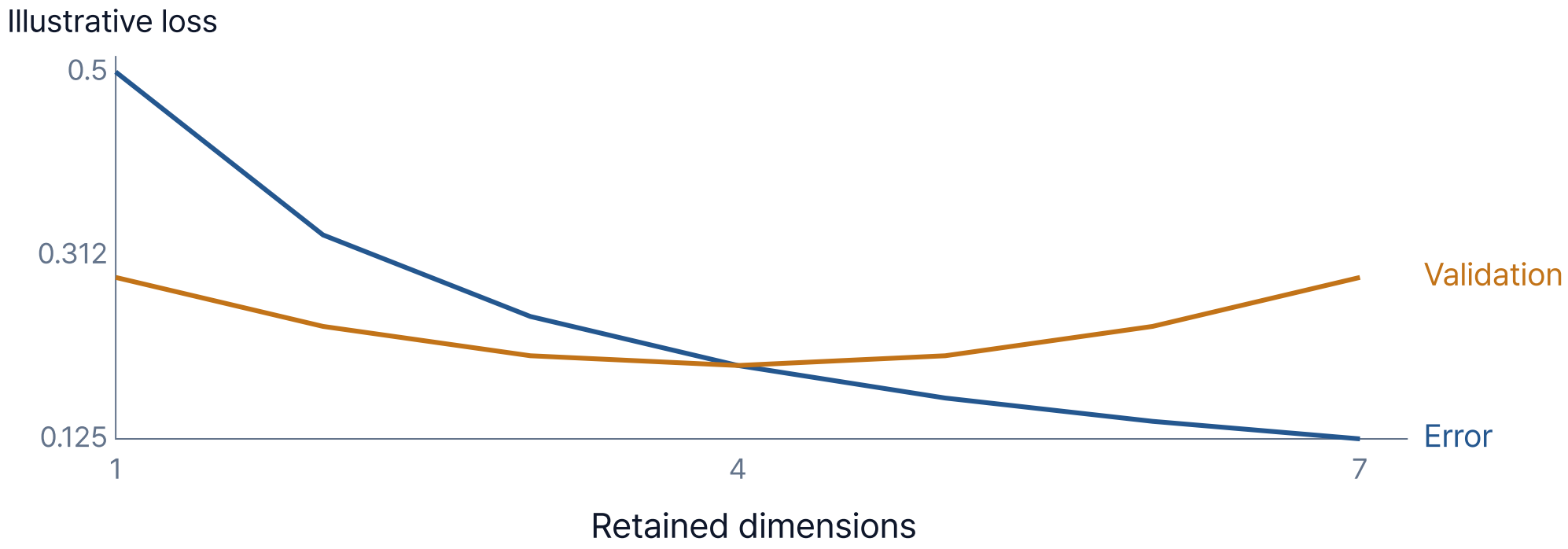
Feature scores must be fitted inside each training fold



Selecting features before splitting leaks label information

Choosing Representation Size

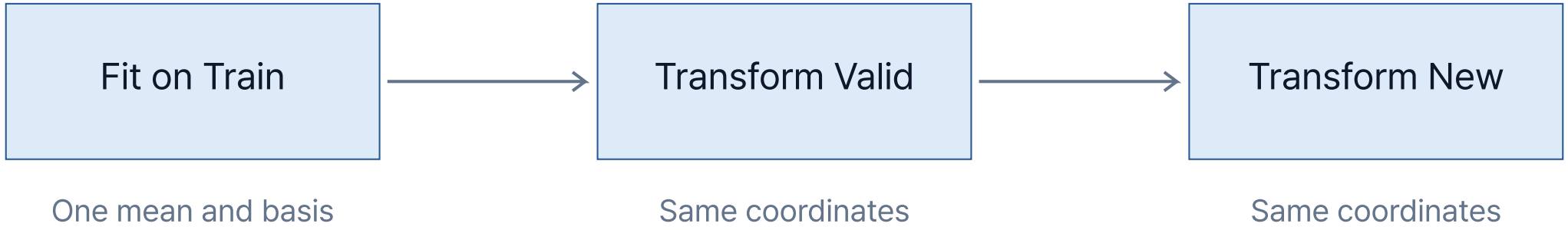
Compare downstream performance and reconstruction cost



More retained variance need not minimize validation loss

Caveat: Preprocessing Shift

Reuse the fitted representation rather than fitting each batch anew



Separate fits produce incompatible coordinate systems

03 · Neighborhood Displays

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Neighborhood Embeddings

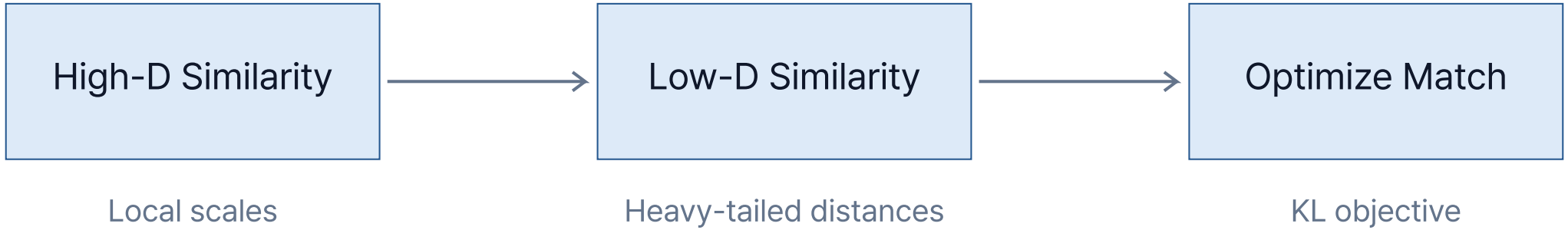
Preserve selected local relationships in a display

	A	B	C
A	0	0.8	0.1
B	0.8	0	0.3
C	0.1	0.3	0

The neighborhood weights define what the display should preserve

t-SNE Affinities

Compare neighborhood probabilities before and after embedding



A point-specific scale sets the local neighborhood

UMAP Neighborhood Graph

Optimize a display from a weighted neighbor graph



Local connectivity

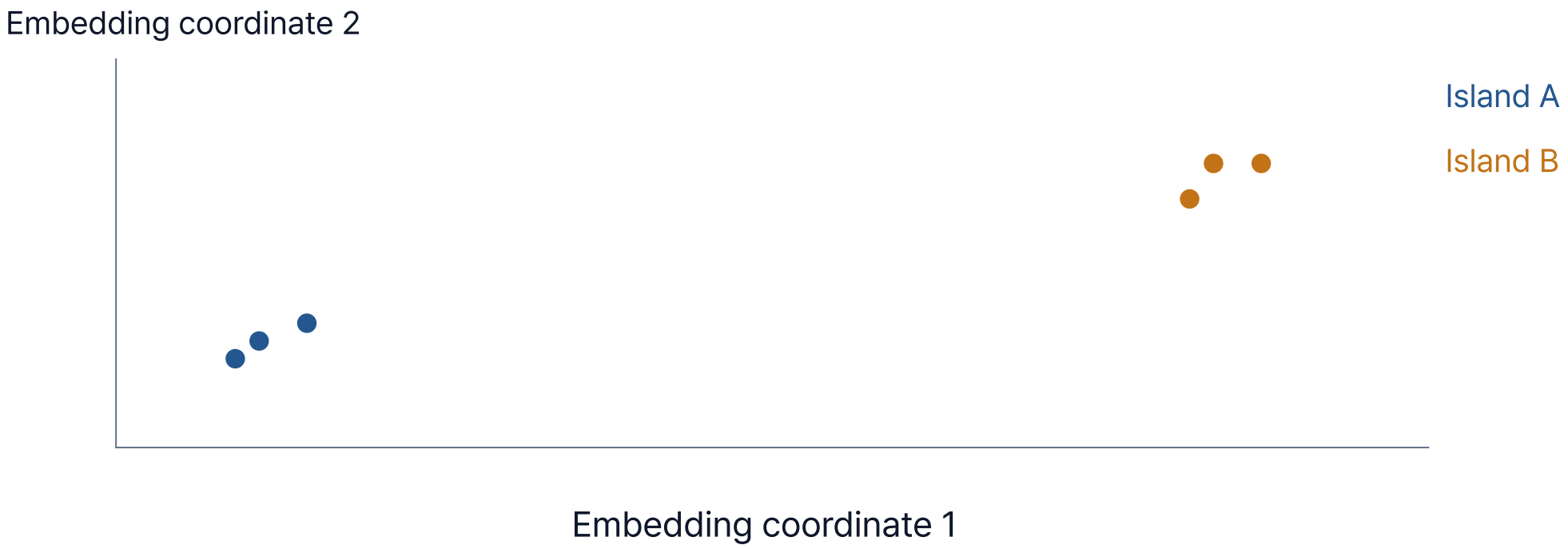
Combine neighborhoods

Optimize layout

The graph construction changes what the layout emphasizes

Caveat: Distances Between Islands

Empty display space is not a calibrated feature distance



Separated islands alone do not measure real-world separation

Caveat: Display Density

A layout can reshape apparent cluster area



The displayed area is not a direct density estimate

Repeated Embedding Checks

Compare seeds and neighborhood settings

Run	Setting
Seed 1	Small neighborhood
Seed 2	Small neighborhood
Seed 1	Large neighborhood

A stable pattern survives relevant layout choices

Reconstruction Error by Direction

A rank-one fit discards the smaller singular direction



The discarded component determines the lost squared variation

Representation Audit

Record the objective before interpreting the coordinates

Question	Evidence
What is preserved?	Variance, labels, or neighbors
What was fitted?	Train mean, basis, graph
What is lost?	Reconstruction or neighbor errors
What is tested?	Validation and stability

Interpretation follows the representation objective

Takeaway: Dimensionality Reduction

01 Match the representation to its objective

02 Fit preprocessing inside training boundaries

03 Treat display geometry as a model output